

Utilities grapple with a multibillion question: How much AI data center power demand is real

Posted by u/Ouaiy • 0 points • 15 comments

Now it gets real.

OpenAI and others have been talking about building gigawatts' worth of data centers. These gigawatts have to come from somewhere, and utility companies have to make the decision now. Suppose they build up generation capacity, and data centers materialize as promised, the utilities stand to earn a fortune. If they don't, the utilities will be left with an enormous investment and no payoff. At this point they need solid numbers, but all that is available is handwaving. Some of the people interviewed are bullish on AI expansion, others are skeptical.

As the article says, solar and wind will be the quickest way to build up power generation capacity, but the administration is hostile to renewable energy. I hope for one of two scenarios: either the lack of power capacity kills off the AI bubble sooner than later; or somehow, extra renewable power generation is built, and when the bubble pops, the country will be left with a surplus of energy which will kill off much of its fossil fuel power generation.

Comments

u/Stoop_Solo • 6 points • 2025-10-19 04:54 UTC

"We're going to have to construct a Dyson sphere so our upcoming planet-sized hyper-cluster can make a video of a horse with Ronald Reagan's face or some shit."

u/Visible_Judge1104 • 1 points • 2025-10-19 04:16 UTC

I would guess they are largely going to make their own power, utility companies are very slow and highly regulated. They don't really have an incentive to not drag their feet. It takes 1 to 2 years just to change their rates and they can only make 10% a year profits. They don't mesh well with the datacenters.

u/Pale_Neighborhood363 • 3 points • 2025-10-18 22:23 UTC

Two errors here. Data Centres need baseline power AND they contract for priority supply. This stuffs the utility companies - as they can ONLY build gas power at the price.

So you get the worst energy mix. Renewables have a twenty plus year cost/return curve as new

infrastructure is needed, Gas power has a four year cost/return curve so the risk is less.

The Green solution is to use coal BUT it will take over twenty years for this to be figured out. Baseload thermal plants + renewables BUT what we get is gas turbines the worst(but most flexible) of both worlds.

u/Ouaiy • 2 points • 2025-10-18 22:26 UTC

I think the argument is that renewables can come online the fastest, especially because of the turbine storage.

u/Pale_Neighborhood363 • 0 points • 2025-10-18 23:00 UTC

The problem is getting the power from the source to the consumer. Renewables need a lot of extra infrastructure (that does not exist). When the extra infrastructure is built out then the argument has some validity.

Renewables are very hard to integrate into existing infrastructure and this is left out of the reports.

The turbine shortage is very short term, a blip. Look at China's power mix build out.

"AI" is not about compute it is about infrastructure usurping. The 'Value' is control of people's essentials - Water, Power, Communications & Finance.

"AI" is to destroy the 'savings' of the middle class, money is set to implode. The question is who gets the 'pot'.

u/woskk • 1 points • 2025-10-20 02:31 UTC

What makes renewables harder to integrate into the grid?

u/Pale_Neighborhood363 • 1 points • 2025-10-20 02:46 UTC

Energy density AND intermittence

Solar needs a lot of extra wiring and converters.

The intermittence is what crashed Spain's grid. Wind is very intermittent.

Basically renewables doubles the 'size' of the grid.

u/Ouaiy • 1 points • 2025-10-19 02:25 UTC

From what I read, turbine shortage is not short term:

[<https://www.bloomberg.com/news/newsletters/2025-10-02/the-gas-turbine-shortage-might-be-a-climate-problem>]

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"Three companies control 70% of the global market for gas turbines and they remain skeptical about whether the recent surge in demand for turbines will last long enough to recoup investments needed to build new manufacturing capacity." Same as with the utility companies: they are

loath to gamble on AI.

Some are expanding its gas turbine production, but it will take a couple of years to get the expanded production going. Right now the wait time is seven years for some types:

[<https://www.utilitydive.com/news/mitsubishi-gas-turbine-manufacturing-capacity-expansion-supply-demand/759371/>]

(<https://www.utilitydive.com/news/mitsubishi-gas-turbine-manufacturing-capacity-expansion-supply-demand/759371/>)

u/Pale_Neighborhood363 • 1 points • 2025-10-19 03:06 UTC

That is marketing NOT fact, the aviation industry can fill the gap for a price... Lots of games here.

Several not Seven, but the demand is equal to renewables as the grid has to match transient capacity. Gas turbine is the cheapest backup short term. This is a distribution not a capacity problem, it is mid size turbines that are relatively scarce as they compensate for lack of infrastructure development.

u/Beginning-Art7858 • 12 points • 2025-10-18 21:46 UTC

Yeah this is the kind of stuff that requires decades to setup. I think the plan is to let the rates go up and claim the market will eventually sort it out.

We are gonna have crazy electric bills nation wide

u/Maximum-Objective-39 • 1 points • 2025-10-19 16:46 UTC

At this rate we're all gonna have to become solar punks just to cook dinner.

u/Beginning-Art7858 • 1 points • 2025-10-19 17:31 UTC

Oddly it would force actual humans to consume less while at the same time so kinda a pro for the environment?

u/Maximum-Objective-39 • 1 points • 2025-10-19 17:38 UTC

30 years from now conservatives will insist this was all part of GENIUS PRESIDENT'S TRUMP'S! Plan to get people to adopt home solar. -sigh-

u/Beginning-Art7858 • 1 points • 2025-10-19 18:02 UTC

Im so sorry that I agree lol

u/LBishop28 • 3 points • 2025-10-19 04:04 UTC

I wonder how much money will be tied up in court battles because communities are starting to say fuck data centers. Nobody wants to subsidize rich companies' energy.